

Indian Statistical Institute

Data Analysis with R and Python

Mid-semester Examination

Bachelor of Statistical Data Science

19 MARCH 2025, 10:30AM – 12:30PM

TOTAL POINTS: 20

INSTRUCTIONS

Write your answers on the question paper. You may use separate sheets for calculations, but submit only the filled in question paper.

Keep your answers brief and to the point. Write only the *answers* to the questions asked. Do **not** write any R code or any other explanation except for question 1(a).

This is a practical examination; you may use laptops or desktops with R and necessary packages pre-installed, as per instructions given earlier. However, you are not allowed to access the internet during the examination.

NAME (IN CAPITAL LETTERS):

ROLL NUMBER:

Question 1: Suppose X is a 2×2 matrix whose entries are independent Bernoulli($\frac{1}{2}$) random variables. Let p denote the probability that X has full rank (i.e. $\text{rank}(X) = 2$).

Further, suppose 75 students in a class replicate this experiment independently of each other. Let Y be the number of cases in which the resulting matrix has full rank, and define

$$q = P(Y < 25).$$

(a) Obtain the value of p . Briefly justify your answer. [3]

(b) Using this value of p , obtain the value of q . Hint: use the R function `pbinom()`. [1]

(c) Obtain an approximate value of q using the Central Limit Theorem. Use continuity correction if appropriate. Hint: use the R function `pnorm()`. [1]

Execute the following code in R to obtain a subset of the **NHANES** data, available in the **NHANES** package. All further questions relate to this subset of data.

```
data(NHANES, package = "NHANES")
dsub <- subset(NHANES, Age >= 25 & Race1 == "White")
```

Question 2:

- (a) How many observations (cases) does **dsub** have? [1]
- (b) The variable **MaritalStatus** has 6 levels. Which level has the maximum frequency? [1]
- (c) The variable **HealthGen** has 5 levels. Which level has the maximum frequency? [1]
- (d) In the cross-tabulation (contingency table) of variables **MaritalStatus** and **HealthGen**, [2]
which has 30 combinations, which combination of levels has the maximum frequency?

Question 3:

- (a) What are the units of the variables **Height** and **Weight**? Hint: Use the help system. [1]
- (b) For how many cases are the values of the variable **Height** available (i.e., not missing)? [1]
- (c) For how many cases are the values of the variable **Weight** available? [1]
- (d) For how many cases are the values of *both* the variables **Height** and **Weight** available? [1]

Question 4:

- (a) What is the regression line obtained by the following call? [1]

```
fm1 <- lm(Weight ~ Height, dsub, subset = Gender == "female")
```

- (b) What is the estimated value of `Weight` according to the model `fm1` for a person whose `Height` is 6 feet? [2]

- (c) What is the regression line obtained by the following call? [1]

```
fm2 <- lm(log(Weight) ~ Height, dsub, subset = Gender == "female")
```

- (d) What is the estimated value of `Weight` according to the model `fm2` for a person whose `Height` is 6 feet? [2]

Good luck!